Using Python on cRIO

(!! To Enable Vibe Coding !!)

Daniel Eaton

Chief Field Applications Engineer

CLA, CLED, 20 years “doing” w/ NI

Daniel.eaton@emerson.com

We are here to help!

- Pull us in to design conversations
- Let us help de-risk projects
- Tell us what you want to learn about next

Assumptions

- cRIO is already:
  - Imaged
  - Has NI DAQmx drivers installed
  - Has NI RIO drivers installed
  - User name and password configured

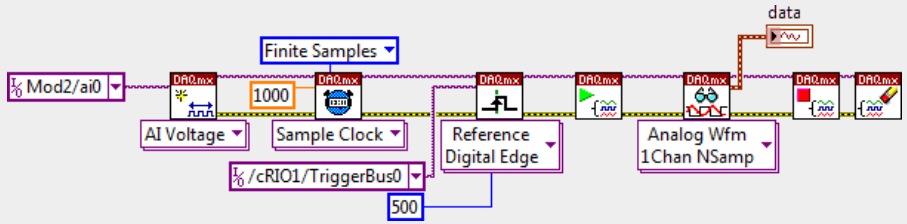
(Start here if you need help: https://www.ni.com/docs/en-US/bundle/crio-904x/page/getting-started-overview.html)

- Your Laptop (Optional but helpful)
  - Has IDE & Python installed
  - Has NI DAQmx installed
  - Has LabVIEW + RT + FPGA installed (IF wanting to program FPGA directly)

Pick the IO API (per slot)

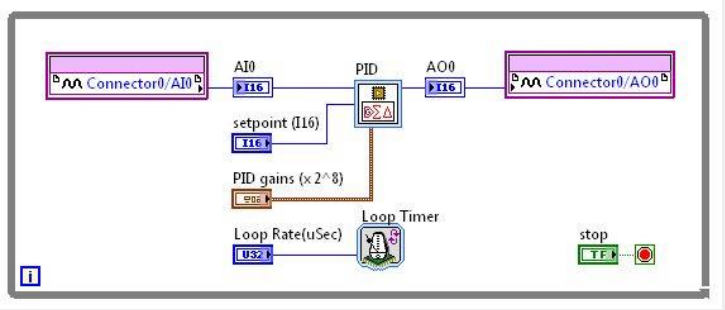
Real-Time (NI-DAQmx)

- Simple API for waveform and control from processor
- Supports control loop rates up to 5 kHz
- All channels in a task are synchronized
- Real-Time uses appropriate driver, such as NI-DAQmx or XNET



LabVIEW FPGA

- Single-point data acquisition and control from LabVIEW FPGA
- Supports control loop rates up to 10s of MHz
- Hardware-level determinism and reliability for control loops
- In-line signal processing for live filtering, calculations, and more



Python APIs for NI DAQmx & FPGA

Documentation = Vibe-able

DAQmx Documentation: https://nidaqmx-python.readthedocs.io/en/stable/

FPGA Documentation: https://nifpga-python.readthedocs.io/en/latest/

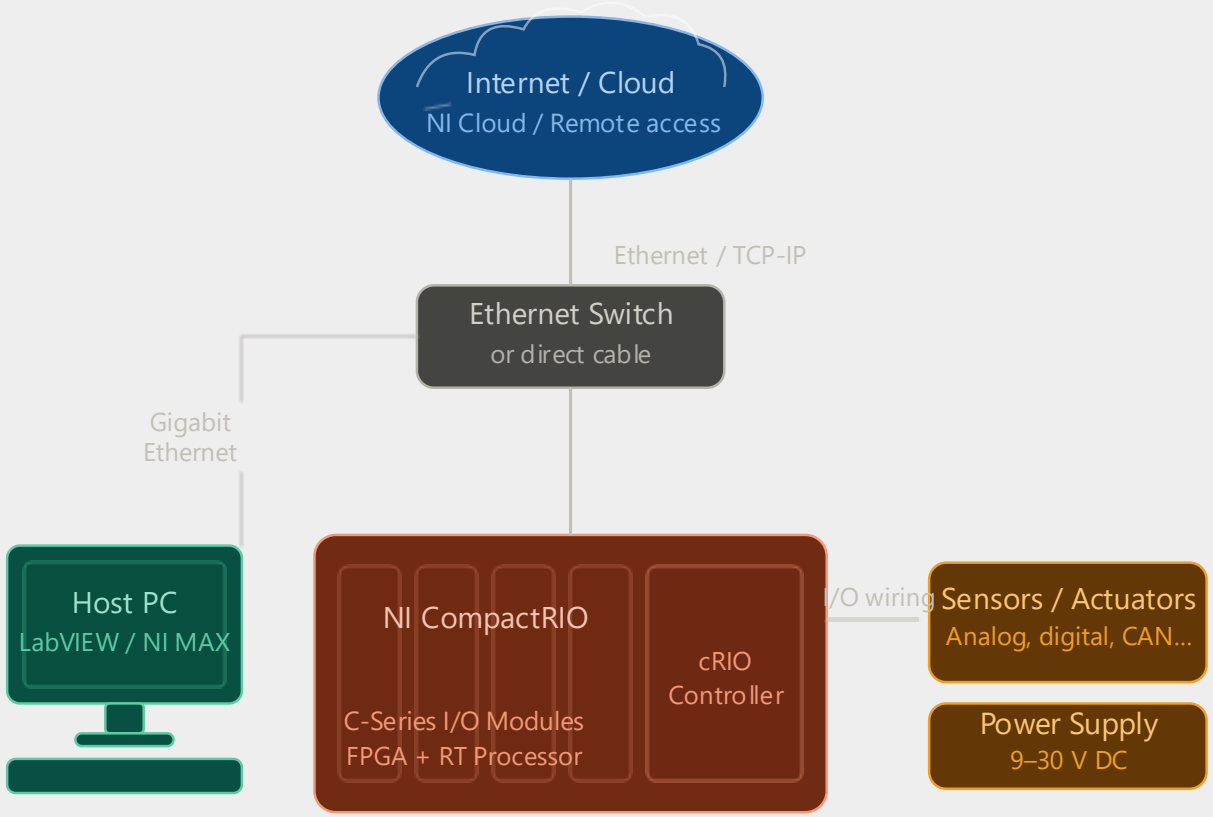
Connect cRIO to Internet & Laptop

Connect your laptop to a switch

Connect your cRIO to a switch

Make sure they are all on the same subnet (255.255.255.0)

Connect the switch to the internet (router, etc)

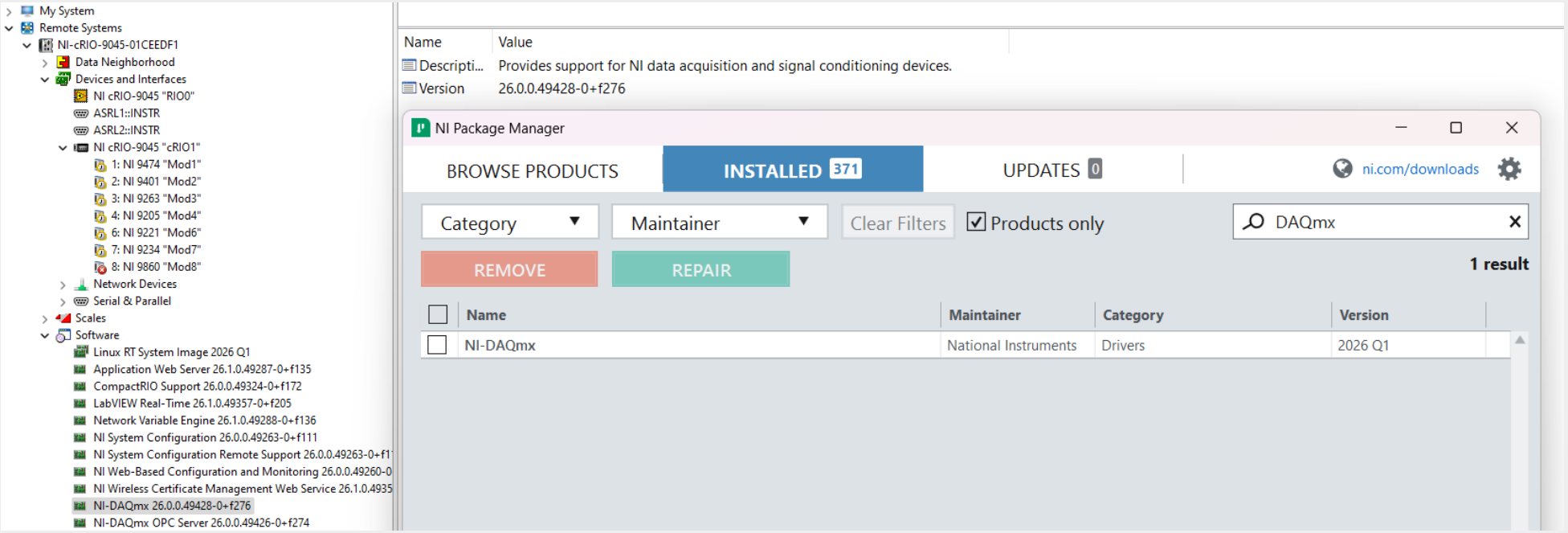


Install Required Drivers

Install NI DAQmx on Host

Install NI DAQmx on cRIO

Make sure the versions match



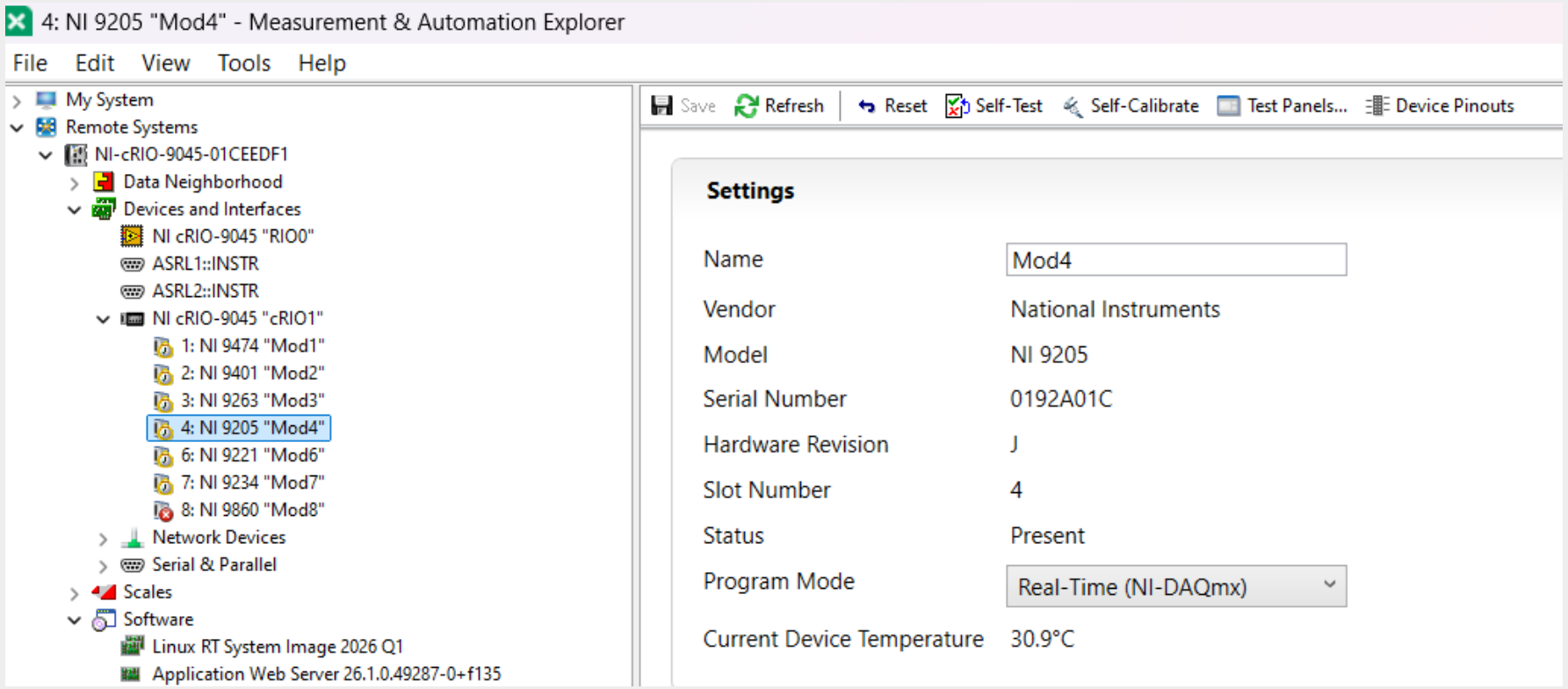
Set Modules to use NI-DAQmx

In Max, under cRIO, per module, set

- Program Mode = NI-DAQmx

Note:

Per module means you can mix NI DAQmx & FPGA access



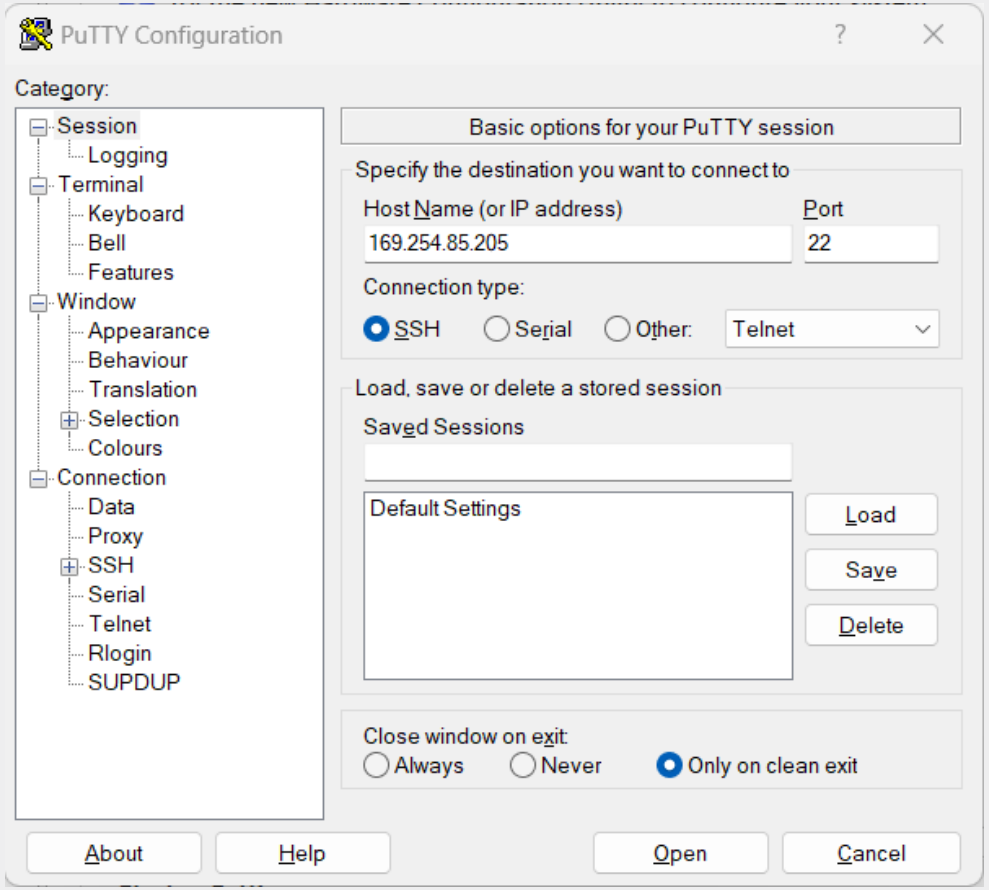
Using PuTTY (SSH) with cRIO

Connect

Set

- Host Name to cRIO IP Address
- Port to 22
- Connection type to SSH

Hit Open

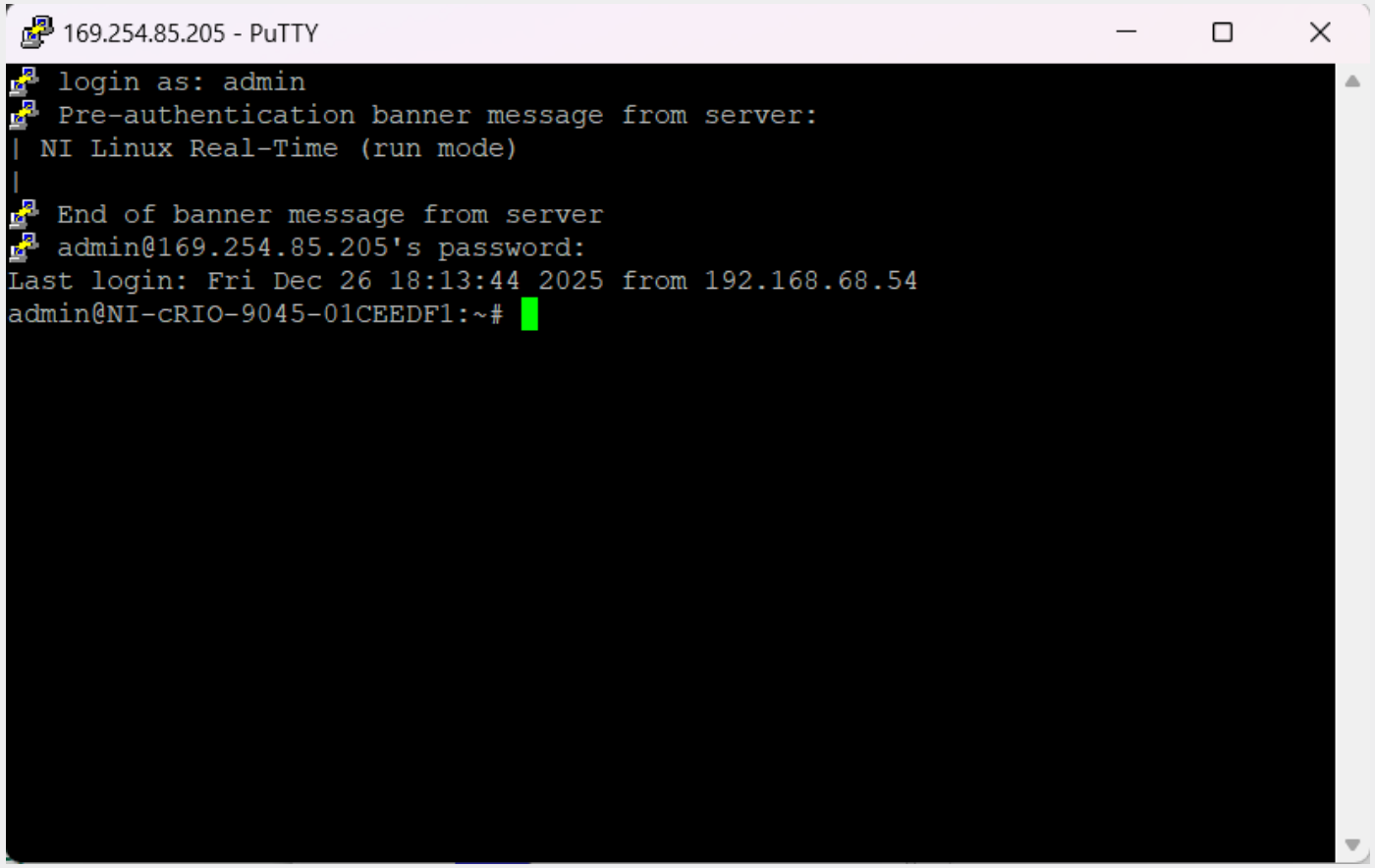


Using PuTTY (SSH) with cRIO

Log In

Enter User Name (& hit enter)

Enter Password (& hit enter)



Using PuTTY (SSH) with cRIO

Install Python

In root, type:

- opkg install python3-pip (Installs Python)

192.168.68.67 - PuTTY

```
login as: admin
Pre-authentication banner message from server:
| NI Linux Real-Time (run mode)
|
End of banner message from server
admin@192.168.68.67's password:
Last login: Tue Mar 17 15:04:44 2026 from 169.254.100.113
admin@NI-cRIO-9045-01CEEDF1:~# opkg install python3-pip
Downloading http://download.ni.com/ni-linux-rt/feeds/2025Q3/x64/main/core2-64/py
thon3-pip-lic_24.0-r0.97_core2-64.ipk.
Downloading http://download.ni.com/ni-linux-rt/feeds/2025Q3/x64/main/core2-64/py
thon3-pip_24.0-r0.97_core2-64.ipk.
Installing python3-pip-lic (24.0) on root
Installing python3-pip (24.0) on root
Configuring python3-pip-lic.
Configuring python3-pip.
admin@NI-cRIO-9045-01CEEDF1:~#
```

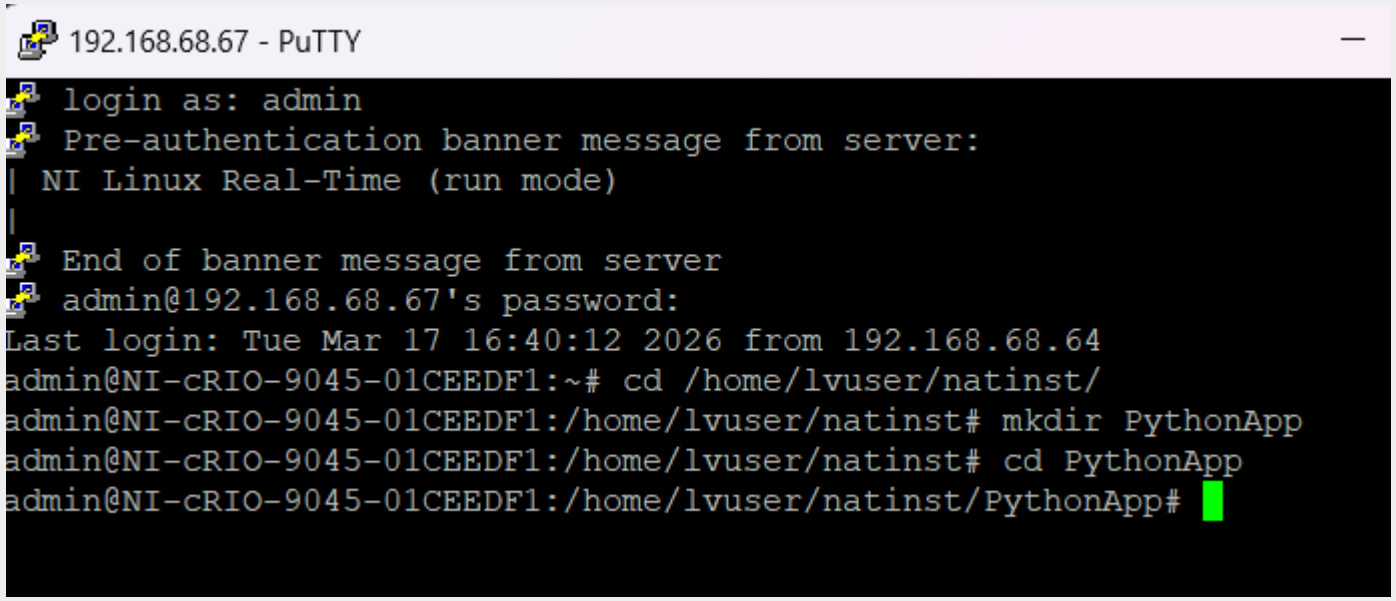
Using PuTTY (SSH) with cRIO

Create the directory

In root, type:

- cd /home/lvuser/natinst/
- mkdir PythonApp
- cd PythonApp

Note: You should be in /home/lvuser/natinst/PythonApp



Using PuTTY (SSH) with cRIO

Create the Virtual Environment & Activate

In /home/lvuser/natinst/PythonApp, type:

- python -m venv venv
- source venv/bin/activate

```
admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# python3 -m venv venv
```

Note: You should now see ((venv))

```
admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# source venv/bin/activate
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp#
```

Using PuTTY (SSH) with cRIO

Upgrade PIP (Package Installer

In ((venv)) /home/lvuser/natinst/PythonApp, type:

- python -m pip install --upgrade pip

```
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# python -m pip install --upgrade pip
Requirement already satisfied: pip in ./venv/lib/python3.12/site-packages (25.0.1)
Collecting pip
  Using cached pip-26.0.1-py3-none-any.whl.metadata (4.7 kB)
Using cached pip-26.0.1-py3-none-any.whl (1.8 MB)
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 25.0.1
    Uninstalling pip-25.0.1:
      Successfully uninstalled pip-25.0.1
Successfully installed pip-26.0.1
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp#
```

Using PuTTY (SSH) with cRIO

Install NI DAQmx Package

In ((venv)) /home/lvuser/natinst/PythonApp, type:

- pip install nidaqmx

```
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# pip install nidaqmx
Collecting nidaqmx
  Using cached nidaqmx-1.4.1-py3-none-any.whl.metadata (17 kB)
Collecting click>=8.0.0 (from nidaqmx)
  Using cached click-8.3.1-py3-none-any.whl.metadata (2.6 kB)
Collecting deprecation>=2.1 (from nidaqmx)
  Using cached deprecation-2.1.0-py2.py3-none-any.whl.metadata (4.6 kB)
Collecting distro>=1.9.0 (from nidaqmx)
  Using cached distro-1.9.0-py3-none-any.whl.metadata (6.8 kB)
Collecting hightime>=0.2.2 (from nidaqmx)
  Using cached hightime-1.0.0-py3-none-any.whl.metadata (2.7 kB)
Collecting nitypes>=1.0.1 (from nidaqmx)
  Using cached nitypes-1.0.1-py3-none-any.whl.metadata (7.0 kB)
Collecting numpy>=1.22 (from nidaqmx)
  Using cached numpy-2.4.3-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.6 kB)
Collecting python-decouple>=3.8 (from nidaqmx)
  Using cached python_decouple-3.8-py3-none-any.whl.metadata (14 kB)
Collecting requests>=2.25.0 (from nidaqmx)
  Using cached requests-2.32.5-py3-none-any.whl.metadata (4.9 kB)
Collecting typing_extensions>=4.0.0 (from nidaqmx)
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Collecting tzlocal<6.0,>=5.0 (from nidaqmx)
  Using cached tzlocal-5.3.1-py3-none-any.whl.metadata (7.6 kB)
Collecting packaging (from deprecation>=2.1->nidaqmx)
  Using cached packaging-26.0-py3-none-any.whl.metadata (3.3 kB)
Collecting charset_normalizer<4,>=2 (from requests>=2.25.0->nidaqmx)
  Using cached charset_normalizer-3.4.6-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.me
tadata (40 kB)
Collecting idna<4,>=2.5 (from requests>=2.25.0->nidaqmx)
```

Using PuTTY (SSH) with cRIO

Install NI FPGA Package

In ((venv)) /home/lvuser/natinst/PythonApp, type:

- pip install nifpga

```
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# pip install nifpga
Collecting nifpga
  Downloading nifpga-22.0.0-py3-none-any.whl.metadata (2.6 kB)
Collecting future (from nifpga)
  Downloading future-1.0.0-py3-none-any.whl.metadata (4.0 kB)
Downloading nifpga-22.0.0-py3-none-any.whl (38 kB)
Downloading future-1.0.0-py3-none-any.whl (491 kB)
Installing collected packages: future, nifpga
Successfully installed future-1.0.0 nifpga-22.0.0
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp#
```

Using FileZilla for File Transfer

(Manual Way)

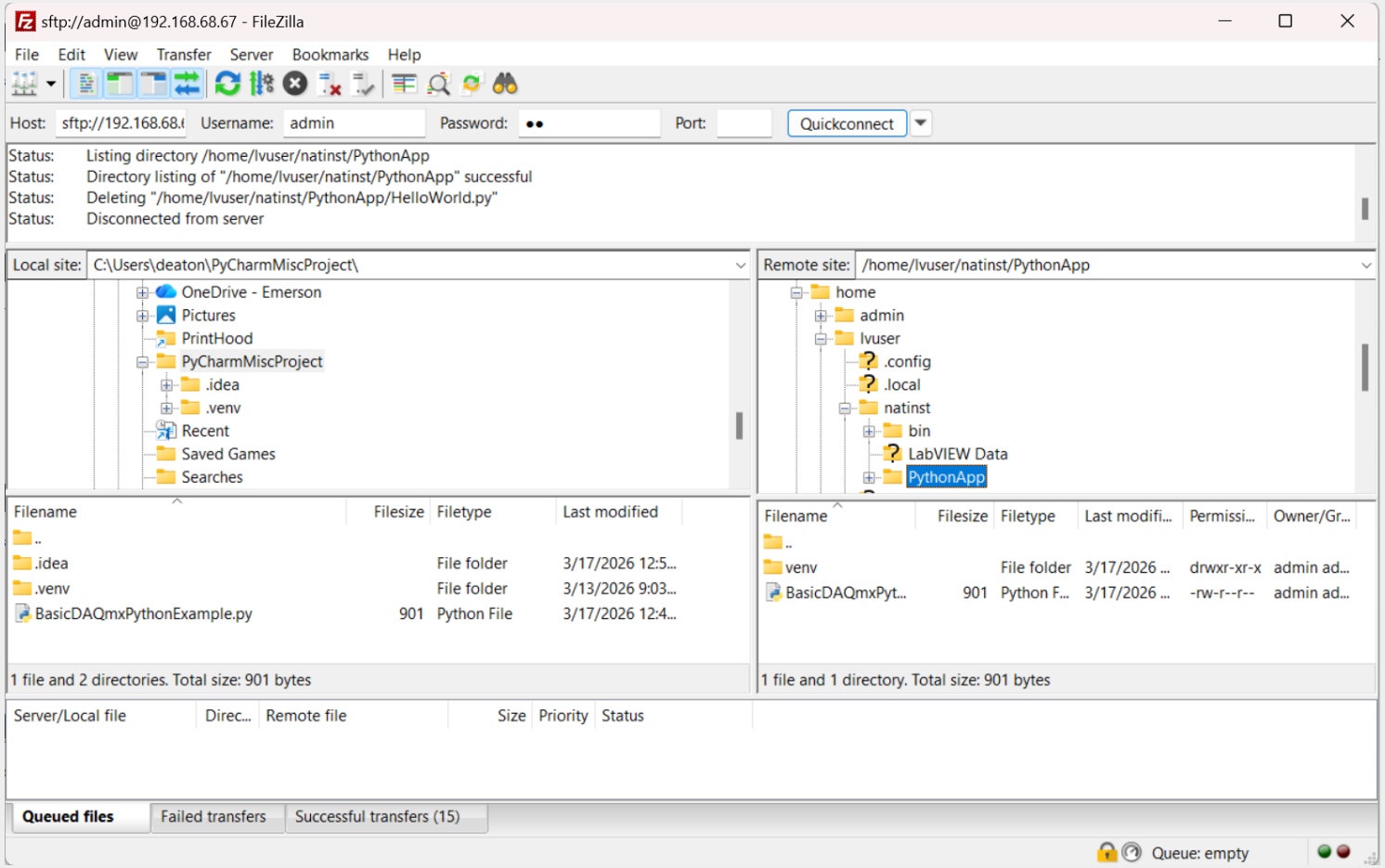
Set

- Host Name to cRIO IP Address
- Enter Username
- Enter Password
- Port to 22

Hit Quick Connect

Move python scripts to:

/home/lvuser/natinst/PythonApp



Run Python Code

(Manual Way)

In ((venv)) /home/lvuser/natinst/PythonApp, type:

- python <scriptname>.py

Using provided example:

- python BasicDAQmxPythonExample.py

192.168.68.67 - PuTTY

```
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# python BasicDAQmxPythonExample.py
DAQmx Version: <function version at 0x7fe2c6a1cae0>
Acquired data: [-0.00210919 -0.00210919 -0.00194503 -0.00145256 -0.00309414 -0.00194503
 -0.0012884  -0.00161672 -0.00194503 -0.00260167 -0.00194503 -0.00161672
 -0.00260167 -0.00194503 -0.00210919 -0.00194503 -0.00227335 -0.00210919
 -0.00178087 -0.00227335 -0.00227335 -0.00227335 -0.0012884  -0.00178087
 -0.00276582 -0.00145256 -0.00161672 -0.00145256 -0.00243751 -0.00145256
 -0.00178087 -0.00243751 -0.00161672 -0.00145256 -0.00260167 -0.00243751
 -0.00194503 -0.00161672 -0.00096008 -0.00227335 -0.00161672 -0.00227335
 -0.00243751 -0.00227335 -0.00276582 -0.00194503 -0.00210919 -0.00210919
 -0.00161672 -0.00194503 -0.00178087 -0.00161672 -0.00210919 -0.00145256
 -0.00227335 -0.00210919 -0.00178087 -0.00096008 -0.00227335 -0.00227335
 -0.00194503 -0.00227335 -0.00260167 -0.00243751 -0.00161672 -0.00096008
 -0.00210919 -0.00243751 -0.00194503 -0.00227335 -0.00178087 -0.00276582
 -0.00145256 -0.00210919 -0.00161672 -0.00210919 -0.00194503 -0.00243751
 -0.00194503 -0.0012884  -0.00194503 -0.00260167 -0.00243751 -0.00161672
 -0.00210919 -0.00227335 -0.00178087 -0.00178087 -0.0012884  -0.00210919
 -0.00243751 -0.0012884  -0.0012884  -0.00243751 -0.00227335 -0.00227335
 -0.00194503 -0.00178087 -0.00178087 -0.00194503]
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp#
```

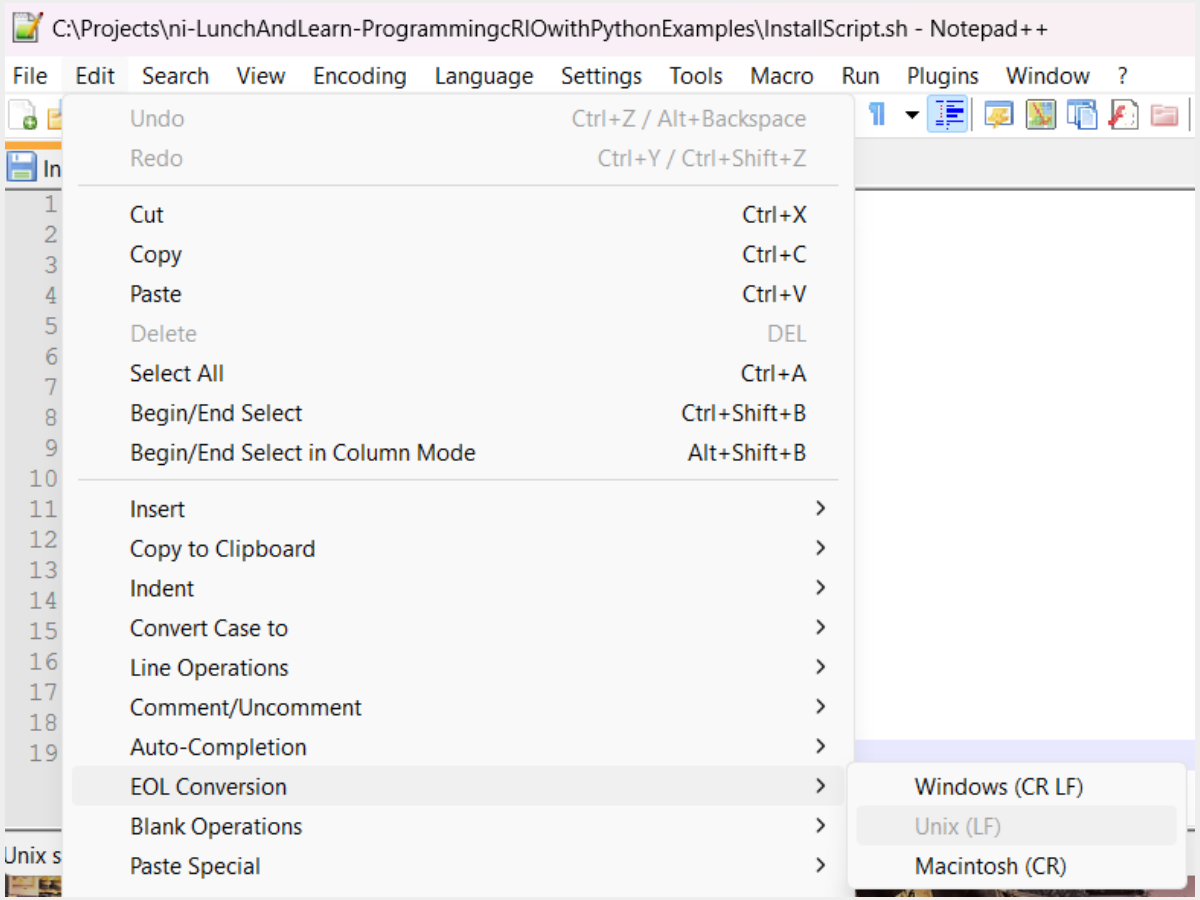
```bash
#!/bin/bash

echo "Installing Python..."
opkg install python3-pip
echo "Moving to User Directory..."
cd '/home/lvuser/natinst/'
echo "Creating New Application Directory..."
mkdir 'PythonApp'
echo "Moving to New Application Directory..."
cd 'PythonApp'
echo "Create Virtual Environment..."
python -m venv venv
echo "Activate Virtual Environment..."
source venv/bin/activate
echo "Upgrading PiP..."
python -m pip install --upgrade pip
echo "Install NI DAQmx..."
pip install nidaqmx
pip install nifpga
```

Demo: Run BASH Script

TIP: Writing a BASH File for NI Linux RT

Extension of .sh

Change EOL Conversion to Unix (LF)



DEMO: Run BASH Script

Review the BASH file (InstallScript.sh)

- Highlight setting up folder structure
- Highlight creating and activating virtual environment
- Highlight installing nidaqmx and nifpga

Use FileZilla to connect to the cRIO (Host = cRIO IP, port 22)

Move InstallScript.sh to the root/default cRIO directory

Use PuTTY to connect to the cRIO (Host = cRIO IP, port 22)

Login to cRIO via SSH (through PuTTY)

Run the script: (Type: bash InstallScript.sh)

See: Videos\Run BASH Script.mp4

```bash
#!/bin/bash

echo "Installing Python..."
opkg install python3-pip
echo "Moving to User Directory..."
cd '/home/lvuser/natinst/'
echo "Creating New Application Directory..."
mkdir 'PythonApp'
echo "Moving to New Application Directory..."
cd 'PythonApp'
echo "Create Virtual Environment..."
python -m venv venv
echo "Activate Virtual Environment..."
source venv/bin/activate
echo "Upgrading PiP..."
python -m pip install --upgrade pip
echo "Install NI DAQmx..."
pip install nidaqmx
pip install nifpga
```

DEMO: Run BASH Script Summary

```bash
#!/bin/bash

echo "Installing Python..."
opkg install python3-pip
echo "Moving to User Directory..."
cd '/home/lvuser/natinst/'
echo "Creating New Application Directory..."
mkdir 'PythonApp'
echo "Moving to New Application Directory..."
cd 'PythonApp'
echo "Create Virtual Environment..."
python -m venv venv
echo "Activate Virtual Environment..."
source venv/bin/activate
echo "Upgrading PiP..."
python -m pip install --upgrade pip
echo "Install NI DAQmx..."
pip install nidaqmx
pip install nifpga
```

Run BASH files

Create a dedicated folder for Python code

Create a virtual environment & activate

Install NI DAQmx and NI FPGA Python packages

Ultimately, cRIO in (NI) Linux (RT)

BasicDAQmxPythonExample.py

```python

with nidaqmx.Task() as task:
    # Add an analog input voltage channel
    task.ai_channels.add_ai_voltage_chan(f"{mod_name}/{chan_name}")

    # Configure simple timing task
    task.timing.cfg_samp_clk_timing(rate=sample_rate, samps_per_chan=num_samples, sample_mode=AcquisitionType.FINITE)

    # Allocate buffer
    data = np.zeros(num_samples, dtype=np.float64)
    reader = AnalogSingleChannelReader(task.in_stream)

    # Start and read data
    task.start()
    reader.read_many_sample(data, number_of_samples_per_channel=num_samples, timeout=10.0)
```

Demo: Python + DAQmx on cRIO

DEMO: Python + DAQmx on cRIO

Review NI DAQmx Python Examples\BasicDAQmxPythonExample.py

- Highlight cRIO hardware (modules being used)
- Highlight import nidaqmx
- Highlight nidaqmx functions

Use FileZilla to connect to the cRIO (Host = cRIO IP, port 22)

- Move BasicDAQmxPythonExample.py to /home/lvuser/natinst/PythonApp

Use PuTTY to connect to the cRIO (Host = cRIO IP, port 22)

- Change Directory: cd '/home/lvuser/natinst/PythonApp'
- Activate Virtual Env: source venv/bin/activate
- Run Python Script: python BasicDAQmxPythonExample.py

See: Videos\Python + DAQmx on cRIO.mp4

192.168.68.67 - PuTTY

```
login as: admin
Pre-authentication banner message from server:
| NI Linux Real-Time (run mode)
|
End of banner message from server
Last login: Wed Mar 18 14:21:58 2026 from 192.168.68.64
admin@NI-cRIO-9045-01CEEDF1:~# cd '/home/lvuser/natinst/PythonApp'
admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# source venv/bin/activate
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# python BasicDAQmxPythonExample.py
BasicDAQmxPythonExample.py is executing on Linux
DAQmx Version: <function version at 0x7f2592ec0ae0>
Acquired data: [ 1.89026870e-04 -1.39289974e-04 -3.03448396e-04 -3.03448396e-04
  2.48684476e-05  2.48684476e-05  2.48684476e-05  2.48684476e-05
  2.48684476e-05 -1.39289974e-04 -1.39289974e-04 -3.03448396e-04
 -1.39289974e-04  1.89026870e-04 -1.39289974e-04 -1.39289974e-04
 -1.39289974e-04  2.48684476e-05  2.48684476e-05  2.48684476e-05
 -3.03448396e-04 -1.39289974e-04 -1.39289974e-04  2.48684476e-05
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  2.48684476e-05 -1.39289974e-04 -1.39289974e-04  1.89026870e-04
  2.48684476e-05  2.48684476e-05 -1.39289974e-04 -1.39289974e-04
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  2.48684476e-05  2.48684476e-05  1.89026870e-04 -1.39289974e-04
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 -3.03448396e-04 -1.39289974e-04  2.48684476e-05 -3.03448396e-04
  2.48684476e-05 -1.39289974e-04  2.48684476e-05 -1.39289974e-04
 -1.39289974e-04  2.48684476e-05 -1.39289974e-04  1.89026870e-04
  2.48684476e-05 -1.39289974e-04 -1.39289974e-04 -1.39289974e-04]
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp#
```
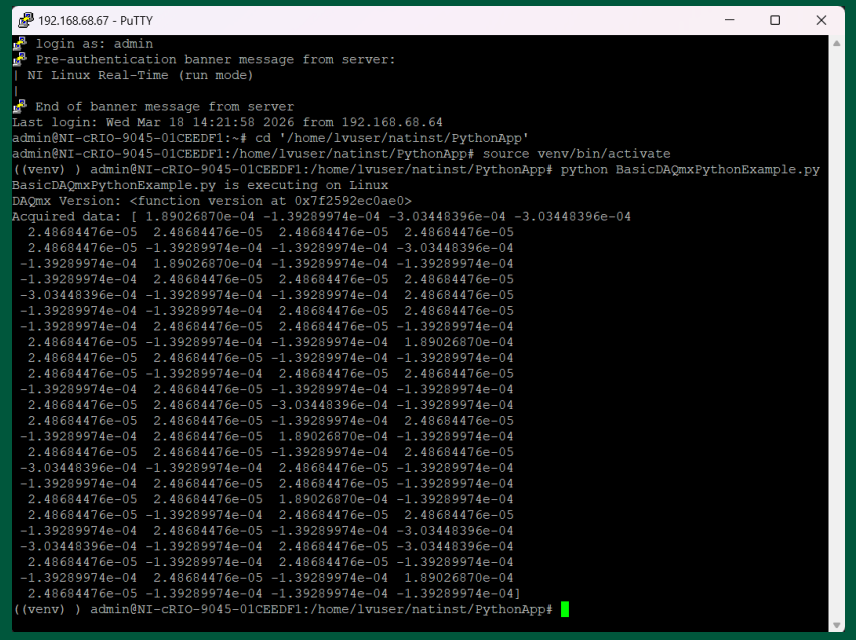
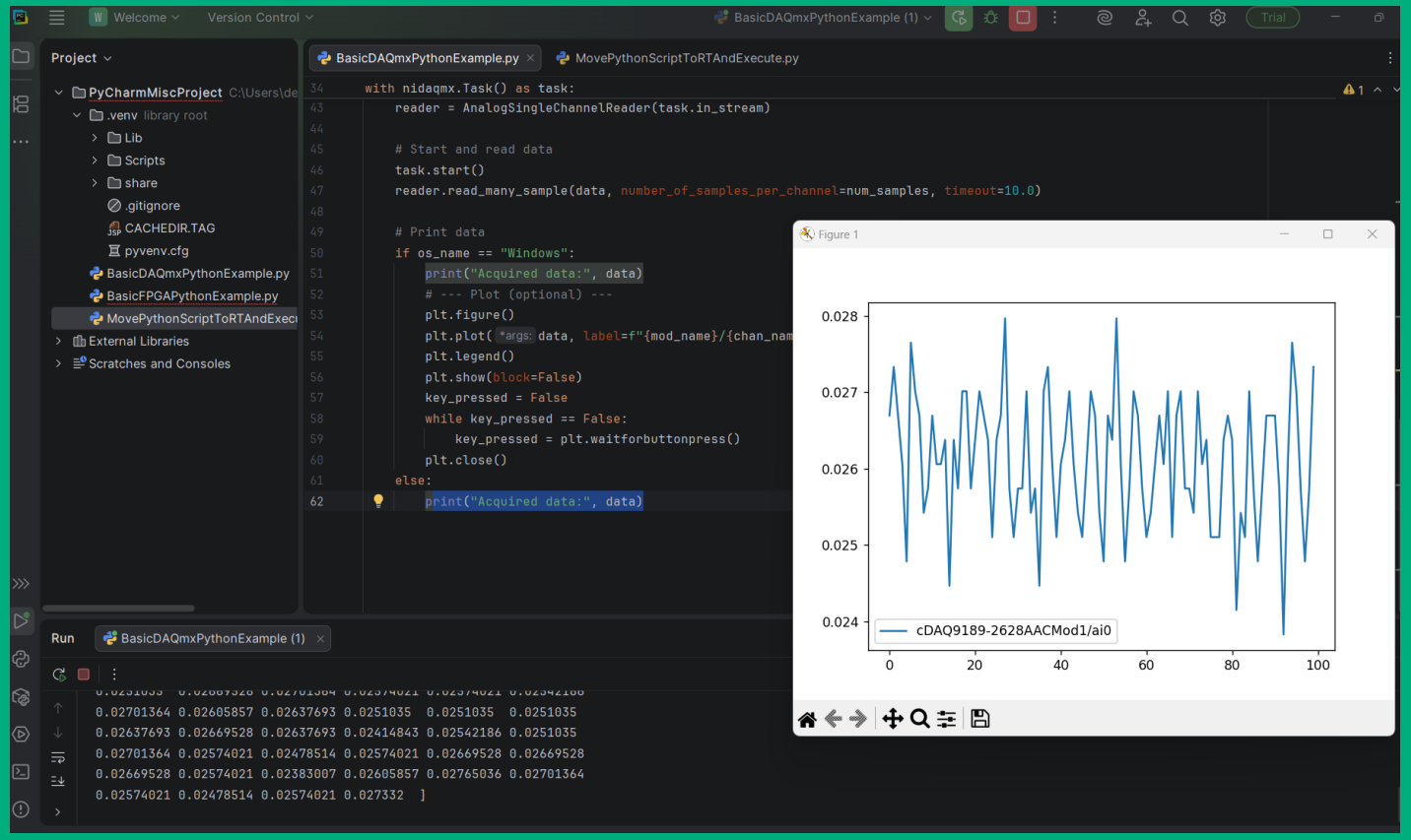

DEMO: Python + DAQmx on cRIO Summary



Run Python on cRIO

Use NI DAQmx API



Demo: Run Python on Windows w/ cDAQ (DAQmx Only)

EMERSON

TIP…Run with cDAQ on Windows First…(DAQmx Only)

Easiest Debug

```python
os_name = platform.system()
if os_name == "Windows":
    import matplotlib.pyplot as plt

#Highlight NI DAQmx works on Windows and on NI Linux RT
print(f"{Path(__file__).name} is executing on {os_name}")
if os_name == "Windows":
    #Set Module Names for the debug system
    mod_name = "cDAQ9189-2628AACMod1"
elif os_name == "Linux":
    #Set Module Names for the cRIO
    mod_name = "Mod4"
```

```python
# Print data
if os_name == "Windows":
    print("Acquired data:", data)
    # --- Plot (optional) ---
    plt.figure()
    plt.plot( *args: data, label=f"{mod_name}/{chan_name}")
    plt.legend()
    plt.show(block=False)
    key_pressed = False
    while key_pressed == False:
        key_pressed = plt.waitforbuttonpress()
    plt.close()
else:
    print("Acquired data:", data)
```

Debugging (DAQmx only)

cRIO is an embedded target (Limited GUI)

Windows is easier for debugging

Build cases for running the same code on Windows as on NI Linux RT

- Import based on OS
- GUI Plotting based on OS

Debug most of the code on Windows

Use a cDAQ 9189 chassis instead of cRIO chassis

- Use same NI DAQmx API
- Use C Series Modules
- Use same wiring

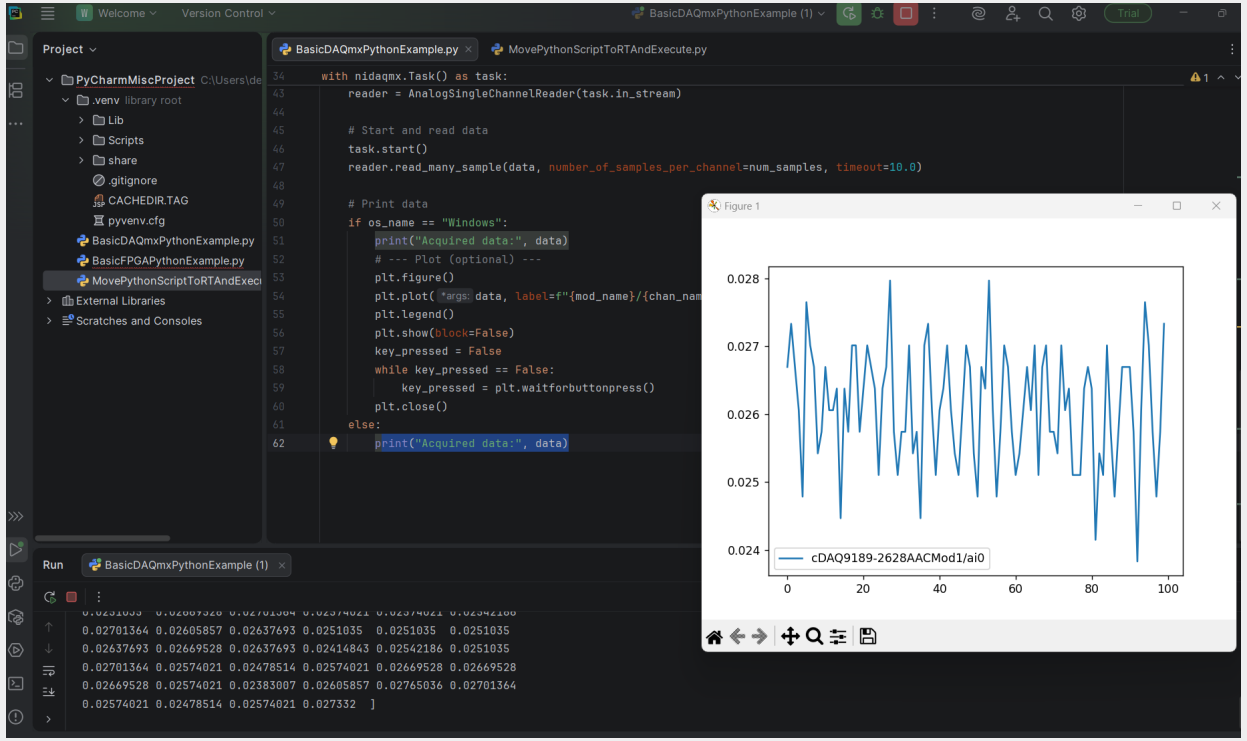
Debugging (DAQmx only)

Review NI DAQmx Python Examples\BasicDAQmxPythonExample.py

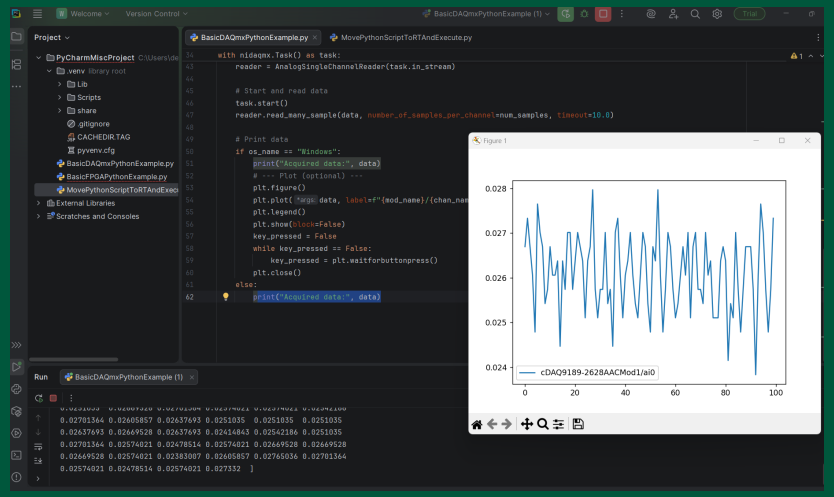
- Highlight OS based cases
  - Import
  - Plotting

Run BasicDAQmxPythonExample.py on Windows

See: Videos\Run Python on Windows with cDAQ_DAQmx Only.mp4



DEMO: Run Python on Windows w/ cDAQ (DAQmx only) Summary



Run the same Python script fully on Windows

Only swap cRIO Chassis for cDAQ Chassis

Configuration changes slightly, main code UNTOUCHED

```
Run    MoveAndExecute

C:\Users\deaton\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\deaton\PyCharmMiscProject\MoveAndExecute.py
Running on: Windows
Remote command: cd /home/lvuser/natinst/PythonApp/ && /home/lvuser/natinst/PythonApp/venv/bin/python BasicDAQmxPythonExample.py
Connecting to 192.168.68.67 using password...
Transferring 'C:\Users\deaton\PyCharmMiscProject\BasicDAQmxPythonExample.py' -> '192.168.68.67:/home/lvuser/natinst/PythonApp//BasicDAQmxPythonExample.py'...
Transfer verified. File size: 1885 bytes
Connected to 192.168.68.67

Changing to directory: /home/lvuser/natinst/PythonApp/
Executing script: BasicDAQmxPythonExample.py

--------------------------------------------------
Output:
  BasicDAQmxPythonExample.py is executing on Linux
  DAQmx Version: <function version at 0x7fa8ce25c680>
  Acquired data: [-0.00227335 -0.00292998 -0.00210919 -0.00161672 -0.00243751 -0.00243751
   -0.00243751 -0.00276582 -0.00227335 -0.00112424 -0.00243751 -0.00178087
   -0.0012884  -0.00227335 -0.00276582 -0.00227335 -0.00161672 -0.00227335
   -0.0012884  -0.00243751 -0.00260167 -0.00260167 -0.00227335 -0.00178087
   -0.00227335 -0.00194503 -0.00194503 -0.00292998 -0.00161672 -0.00276582
   -0.00178087 -0.00178087 -0.00210919 -0.00194503 -0.00194503 -0.00227335
   -0.00178087 -0.00227335 -0.00161672 -0.00178087 -0.00194503 -0.00243751
   -0.00260167 -0.00260167 -0.00227335 -0.00210919 -0.00145256 -0.0012884
   -0.0012884  -0.00178087 -0.00194503 -0.00210919 -0.00178087 -0.00178087
```

Demo: Automate Remote Workflow

Remote Development Options

Windows → NI Linux RT

Some IDEs allow automatic file sync and SSH connection

Use a python script to transfer and run your application

Demo: Automate Remote Workflow

Review NI DAQmx Python Examples\MovePythonScriptToRTAndExecute.py

- Highlight paramiko package
- Highlight SSH
- Highlight sFTP
- Highlight remote command
- Highlight streaming back response

In FileZilla

- Connect to cRIO
- GoTo: /home/lvuser/natinst/PythonApp/
- Delete file BasicDAQmxPythonExample.py

Run MovePythonScriptToRTAndExecute.py

- Show Result
- Show File back on cRIO

```
Run    MoveAndExecute
C:\Users\deaton\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\deaton\PyCharmMiscProject\MoveAndExecute.py
Running on: Windows
Remote command: cd /home/lvuser/natinst/PythonApp/ && /home/lvuser/natinst/PythonApp/venv/bin/python BasicDAQmxPythonExample.py
Connecting to 192.168.68.67 using password...
Transferring 'C:\Users\deaton\PyCharmMiscProject\BasicDAQmxPythonExample.py' -> '192.168.68.67:/home/lvuser/natinst/PythonApp//BasicDAQmxPythonExample.py'...
Transfer verified. File size: 1885 bytes
Connected to 192.168.68.67

Changing to directory: /home/lvuser/natinst/PythonApp/
Executing script: BasicDAQmxPythonExample.py

--------------------------------------------------
Output:
  BasicDAQmxPythonExample.py is executing on Linux
  DAQmx Version: <function version at 0x7fa8ce25c680>
  Acquired data: [-0.00227335 -0.00292998 -0.00210919 -0.00161672 -0.00243751 -0.00243751
   -0.00243751 -0.00276582 -0.00227335 -0.00112424 -0.00243751 -0.00178087
   -0.0012884  -0.00227335 -0.00276582 -0.00227335 -0.00161672 -0.00227335
   -0.0012884  -0.00243751 -0.00260167 -0.00260167 -0.00227335 -0.00178087
   -0.00227335 -0.00194503 -0.00194503 -0.00292998 -0.00161672 -0.00276582
   -0.00178087 -0.00178087 -0.00210919 -0.00194503 -0.00194503 -0.00227335
   -0.00178087 -0.00227335 -0.00161672 -0.00178087 -0.00194503 -0.00243751
   -0.00260167 -0.00260167 -0.00227335 -0.00210919 -0.00145256 -0.0012884
   -0.0012884  -0.00178087 -0.00194503 -0.00210919 -0.00178087 -0.00178087
```

See: Videos\Automate Remote Workflow.mp4

DEMO: Automate Remote Workflow Summary

```
Run    MoveAndExecute

C:\Users\deaton\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\deaton\PyCharmMiscProject\MoveAndExecute.py
Running on: Windows
Remote command: cd /home/lvuser/natinst/PythonApp/ && /home/lvuser/natinst/PythonApp/venv/bin/python BasicDAQmxPythonExample.py
Connecting to 192.168.68.67 using password...
Transferring 'C:\Users\deaton\PyCharmMiscProject\BasicDAQmxPythonExample.py' -> '192.168.68.67:/home/lvuser/natinst/PythonApp//BasicDAQmxPythonExample.py'...
Transfer verified. File size: 1885 bytes
Connected to 192.168.68.67

Changing to directory: /home/lvuser/natinst/PythonApp/
Executing script: BasicDAQmxPythonExample.py

--------------------------------------------------
Output:
  BasicDAQmxPythonExample.py is executing on Linux
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  Acquired data: [-0.00227335 -0.00292998 -0.00210919 -0.00161672 -0.00243751 -0.00243751
   -0.00243751 -0.00276582 -0.00227335 -0.00112424 -0.00243751 -0.00178087
   -0.0012884  -0.00227335 -0.00276582 -0.00227335 -0.00161672 -0.00227335
   -0.0012884  -0.00243751 -0.00260167 -0.00260167 -0.00227335 -0.00178087
   -0.00227335 -0.00194503 -0.00194503 -0.00292998 -0.00161672 -0.00276582
   -0.00178087 -0.00178087 -0.00210919 -0.00194503 -0.00194503 -0.00227335
   -0.00178087 -0.00227335 -0.00161672 -0.00178087 -0.00194503 -0.00243751
   -0.00260167 -0.00260167 -0.00227335 -0.00210919 -0.00145256 -0.0012884
   -0.0012884  -0.00178087 -0.00194503 -0.00210919 -0.00178087 -0.00178087
```
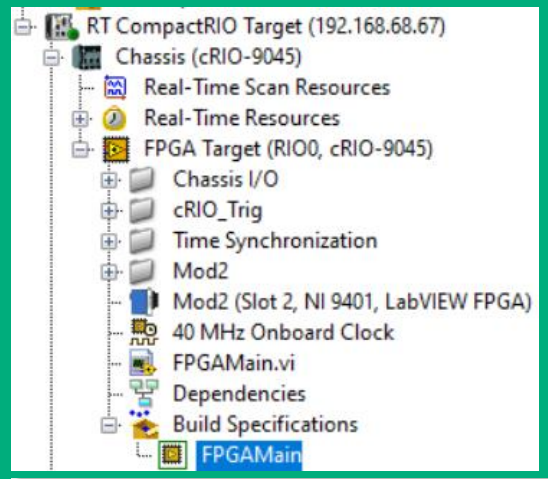

Automatically:

- Move Files to RT
- Run Script on RT
- All Through Windows IDE



192.168.68.67 - PuTTY

```
login as: admin
Pre-authentication banner message from server:
| NI Linux Real-Time (run mode)
|
End of banner message from server
admin@192.168.68.67's password:
Last login: Wed Mar 18 17:32:24 2026 from 192.168.68.64
admin@NI-cRIO-9045-01CEEDF1:~# cd '/home/lvuser/natinst/PythonApp'
admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# source venv/bin/activate
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# python BasicFPGAPythonExample.py
Connecting to FPGA: RIO0
Bitfile: /home/lvuser/natinst/PythonApp/FPGAMainBitfile.lvbitx
Press Ctrl+C to stop.

FPGA is running.

Timestamp Loop#  Count          Simulated Mod2 DIO0
-------------------------------------------------------
17:59:59  1      Count = 1     Simulated Mod2 DIO0 = True
17:59:59  2      Count = 1     Simulated Mod2 DIO0 = False
17:59:59  3      Count = 2     Simulated Mod2 DIO0 = True
17:59:59  4      Count = 2     Simulated Mod2 DIO0 = False
17:59:59  5      Count = 3     Simulated Mod2 DIO0 = True
17:59:59  6      Count = 3     Simulated Mod2 DIO0 = False
17:59:59  7      Count = 4     Simulated Mod2 DIO0 = True
17:59:59  8      Count = 4     Simulated Mod2 DIO0 = False
18:00:00  9      Count = 5     Simulated Mod2 DIO0 = True
18:00:00  10     Count = 5     Simulated Mod2 DIO0 = False
18:00:00  11     Count = 6     Simulated Mod2 DIO0 = True
18:00:00  12     Count = 6     Simulated Mod2 DIO0 = False
18:00:00  13     Count = 7     Simulated Mod2 DIO0 = True
18:00:00  14     Count = 7     Simulated Mod2 DIO0 = False
```

Demo: Python + FPGA on cRIO

Demo: Python + FPGA on cRIO

Prerequisites

Create a LabVIEW Project with your cRIO

Add the FPGA Target

Move Only Desired Modules under the FPGA (PER MODULE)

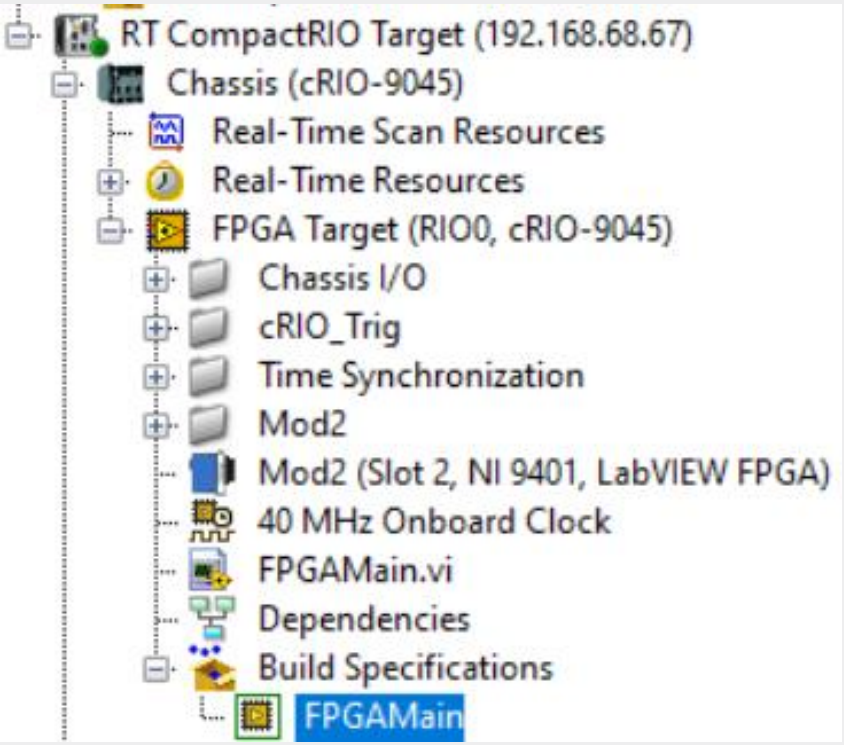
Create your LabVIEW FPGA Program

(Test your Code in Simulated Mode)

Compile your program into a bitfile

(Test your Code w/ Hardware)

Then, write your Python…



Demo: Automate Remote Workflow

Review NI FPGA Python Examples\BasicFPGAPythonExample.py

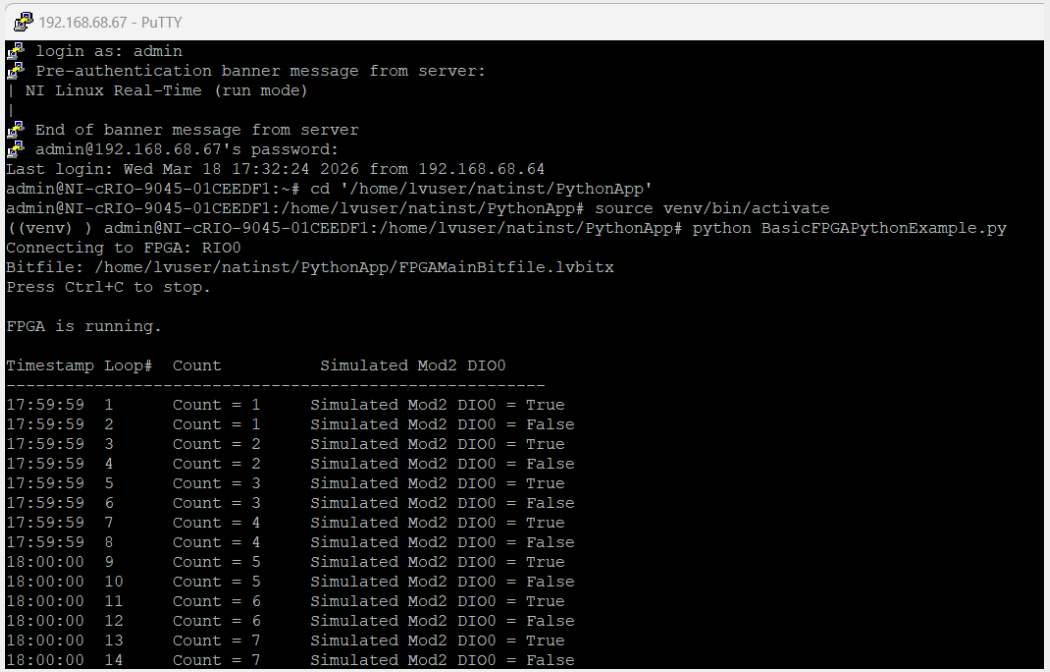
- Highlight import nifpga
- Highlight creating nifpga session
- Highlight starting/reseting fpga
- Highlight reading indicators /writing controls

Use FileZilla to connect to the cRIO (Host = cRIO IP, port 22)

- Move BasicFPGAPythonExample.py to /home/lvuser/natinst/PythonApp

Use PuTTY to connect to the cRIO (Host = cRIO IP, port 22)

- Change Directory: cd '/home/lvuser/natinst/PythonApp’
- Activate Virtual Env: source venv/bin/activate
- Run Python Script: python BasicFPGAPythonExample.py



See: Videos\Python + FPGA on cRIO.mp4

DEMO: Automate Remote Workflow Summary

Use the cRIO FPGA in Python (PER MODULE)

192.168.68.67 - PuTTY

```
login as: admin
Pre-authentication banner message from server:
| NI Linux Real-Time (run mode)
|
End of banner message from server
admin@192.168.68.67's password:
Last login: Wed Mar 18 17:32:24 2026 from 192.168.68.64
admin@NI-cRIO-9045-01CEEDF1:~# cd '/home/lvuser/natinst/PythonApp'
admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# source venv/bin/activate
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# python BasicFPGAPythonExample.py
Connecting to FPGA: RIO0
Bitfile: /home/lvuser/natinst/PythonApp/FPGAMainBitfile.lvbitx
Press Ctrl+C to stop.

FPGA is running.

Timestamp Loop#  Count          Simulated Mod2 DIO0
-------------------------------------------------------
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18:00:00  13     Count = 7     Simulated Mod2 DIO0 = True
18:00:00  14     Count = 7     Simulated Mod2 DIO0 = False
```

And Now…

Go and Vibe!!!

Questions?

Daniel Eaton

Chief Field Applications Engineer

CLA, CLED, 20 years “doing” w/ NI

Daniel.eaton@emerson.com

We are here to help!

- Pull us in to design conversations
- Let us help de-risk projects
- Tell us what you want to learn about next